DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE, CHECK

AIM:

To add the constraints like primary key, foreign key, unique key and check using DDL commands.

Description:

PRIMARY KEY:

The PRIMARY KEY constraint uniquely identifies each record in a database table.

Primary keys must contain UNIQUE values, and cannot contain NULL values.

A table can have only one primary key, which may consist of single or multiple fields.

FOREIGN KEY:

- ➢ A FOREIGN KEY is a key used to link two tables together.
- ➢ A FOREIGN KEY is a field (or collection of fields) in one table that refers to the PRIMARY KEY in another table.
- ➢ The table containing the foreign key is called the child table, and the table containing the candidate key is called the referenced or parent table.

UNIQUE Constraint:

- ➢ The UNIQUE constraint ensures that all values in a column are different.
- ➢ Both the UNIQUE and PRIMARY KEY constraints provide a guarantee for uniqueness for a column or set of columns.
- ➢ A PRIMARY KEY constraint automatically has a UNIQUE constraint.
- ➢ However, you can have many UNIQUE constraints per table, but only one PRIMARY KEY constraint per table.

CHECK Constraint:

➢ The CHECK constraint is used to limit the value range that can be placed in a column ➢ If you define a CHECK constraint on a single column it allows only certain values for this column. ➢ If you define a CHECK constraint on a table it can limit the values in certain columns based on values in other columns in the row.

PRIMARY:

ALTER TABLE table_name

ADD PRIMARY KEY(primary_key_column);

FOREIGN KEY:

```
ALTER TABLE table_name
ADD CONSTRAINT constraint_name
FOREIGN KEY foreign_key_name (columns)
REFERENCES parent_table(columns)
ON DELETE action
ON UPDATE action
```

UNIQUE:

```
CREATE TABLE table_1(
...
column_name_1 data_type,
...
UNIQUE(column_name_1)
);
```

CHECK

```
CREATE TABLE IF NOT EXISTS parts (
part_no VARCHAR(18) PRIMARY KEY,
description VARCHAR(40),
cost DECIMAL(10 , 2 ) NOT NULL CHECK(cost > 0), price DECIMAL (10,2) NOT
NULL
);
```

<u>Questions:</u>

1) Alter the table STUDENT2 with following structure.

#	Column Name	Constraints
	1RegNo	PRIMARY KEY

2) Alter the table name FACULTY with following structure.

#	Column Name	Constraints
1	FacNo	PRIMARY KEY
2	Gender	CHECK ‘M’ or ‘F’

3) Alter the table name DEPARTMENT with following structure.

#	Column Name	Constraints

1	DeptNo	PRIMARY KEY

4) Alter the table name COURSE with following structure.

#	Column Name	Constraints
1	CourseNo	PRIMARY KEY
2	SemNo	1 to 6

5) Alter the table name STUDENT2 with following structure.

#	Column Name	Constraints
1	S1Name	UNIQUE KEY

6) After the STUDENT2 & STUDENT3 tables are successfully created, test if you can add a constraint FOREIGN KEY to the RegNo of this table.

OUTPUTS:

1)

```
mysql> desc student2;
+--------+-------------+------+-----+---------+-------+
| Field  | Type        | Null | Key | Default | Extra |
+--------+-------------+------+-----+---------+-------+
| RegNo  | int         | NO   | PRI | NULL    |       |
| S1Name | varchar(30) | YES  |     | NULL    |       |
| Dept   | varchar(20) | YES  |     | NULL    |       |
+--------+-------------+------+-----+---------+-------+
3 rows in set (0.018 sec)
```

2)

```
ysql> desc faculty;
-------------+----------+------+-----+---------+-------+
 Field       | Type     | Null | Key | Default | Extra |
-------------+----------+------+-----+---------+-------+
 facregno    | int      | NO   | PRI | NULL    |       |
 facname     | char(12) | YES  |     | NULL    |       |
 facgender   | char(3)  | YES  |     | NULL    |       |
 dob         | int      | YES  |     | NULL    |       |
 facmobileno | int      | YES  |     | NULL    |       |
 faccity     | char(10) | YES  |     | NULL    |       |
-------------+----------+------+-----+---------+-------+
 rows in set (0.021 sec)
```

3)

```
mysql> alter table department add primary key(deptno);
Query OK, 0 rows affected (0.11 sec)
```

```
mysql> desc department;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| DeptNo   | int         | NO   | PRI | NULL    |       |
| DeptName | varchar(20) | YES  |     | NULL    |       |
| DeptHead | varchar(20) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
3 rows in set (0.021 sec)
```

4)

```
mysql> desc course;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| CourseNo   | varchar(5)  | NO   | PRI | NULL    |       |
| CourseDesc | varchar(30) | YES  |     | NULL    |       |
| CourseType | char(1)     | YES  |     | NULL    |       |
| SemNo      | int         | YES  |     | NULL    |       |
| HallNo     | varchar(5)  | YES  |     | NULL    |       |
| FacNo      | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
6 rows in set (0.021 sec)
```

5)

```
mysql> desc student2;
+----------+-------------+------+-----+---------+-------+
| Field    | Type        | Null | Key | Default | Extra |
+----------+-------------+------+-----+---------+-------+
| RegNo    | int         | NO   | PRI | NULL    |       |
| S1Name   | varchar(20) | YES  | UNI | NULL    |       |
| Age      | int         | YES  |     | NULL    |       |
| MobileNo | bigint      | YES  |     | NULL    |       |
| Address  | varchar(30) | YES  |     | NULL    |       |
+----------+-------------+------+-----+---------+-------+
5 rows in set (0.023 sec)
```

6)

```
ysql> desc student3;
-------+------+------+-----+---------+-------+
 Field | Type | Null | Key | Default | Extra |
-------+------+------+-----+---------+-------+
 RegNo | int  | YES  | MUL | NULL    |       |
 Marks | int  | YES  |     | NULL    |       |
-------+------+------+-----+---------+-------+
 rows in set (0.024 sec)
```

Result:

DDL Commands with Primary, Foreign, Unique, Check constraints are updated and verified.